OSI Model vs TCP/IP Model

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	OSI Model Layers	Function	TCP/IP Model Layers	PDUs	Hardware	Protocols
7.	Application	Closest to the end user. This is the layer through which the application and the user communicate. For communication between web browsers and web server, application-specific protocols such as HTTP (Hyper Text Transfer Protocol) are utilized at this layer.	Application	Data	Gateways, Proxy Servers, Load Balancers PCs, mobile phones	DNS, FTP, SNMP, DHCP, SSH, SMTP POP3, LDAP, SMB, SSL, TLS NetBIOS HTTPS, HTTP, FTP, NFS,NTP, Telnet, IMAP, SSL, AFP, NetBIOS, RPC, SMB
6.	Presentation	This layer formats the data so that it may be understood by the receiving application. This layer can also encrypt data as it is sent and decrypt it as it is received, ensuring that only the intended recipient can read it.				
5.	Session	This layer controls host-to-host communication (sessions). It creates, manages, and destroys connections between a local application (such as your web browser) and a remote application (for example, youtube).				
4.	Transport	To ensure that no data is lost, the transport layer is employed for error handling and sequencing. This layer also provides host-to-host communication also know as end-to-end communication.	Transport	Segment		TCP, UDP, RTP, SCTP,DCCP
3.	Network	The Network layer connects end hosts on different networks (i.e outside of your LAN). This layer handles logical addressing using IP addresses.	Internet	Packet	Routers, Layer 3 Switches, Brouters	ICMP, IGMP, IP,IPsec, NAT
2.	Data Link	This layer facilitates node-to-node communication and data transfer (for example, pc to switch, switch to router and router to router). The physical address (MAC Address) is appended to the data at this layer, this includes the source and destination MAC addresses.	Network Acces Or Link Layer	Frame	Switches, Bridges, WiFi Access Points	ARP, Ethernet, Token Ring, PPP, ATM, SLIP, Wi-Fi (IEEE 802.11), Frame relay, MAC, PPP, LLDP, L2TP, VLAN, VTP, Bluetooth, ISDN
1.	Physical	The physical layer is the OSI model's bottom layer. It specifies the physical properties of a medium that is used to carry data between devices. For example, Voltage levels, maximum transmission distances, physical connectors, and so forth. Digital bits are transformed to electrical signals for wired connections and radio signals for wireless transmission at this layer.		Bits	Network Cables (e.g ethernet, fiber, copper) Hubs, Repeaters, Network Interface Cards (NICs)	